

$\sigma(z) = \frac{1}{1+e^{-z}}$ $\rightarrow$ loss fn. $\rightarrow$ binary cross entropy
 $L = -Y_i \log \hat{Y}_i + (1-Y_i) \log(1-\hat{Y}_i)$
Logistic Regression
 SGD: Stochastic Gradient Descent $\rightarrow$ $\hat{p}$ (Probability)

multiclass
 Activation $\rightarrow$ Softmax
 loss fn $\rightarrow$ categorical cross entropy

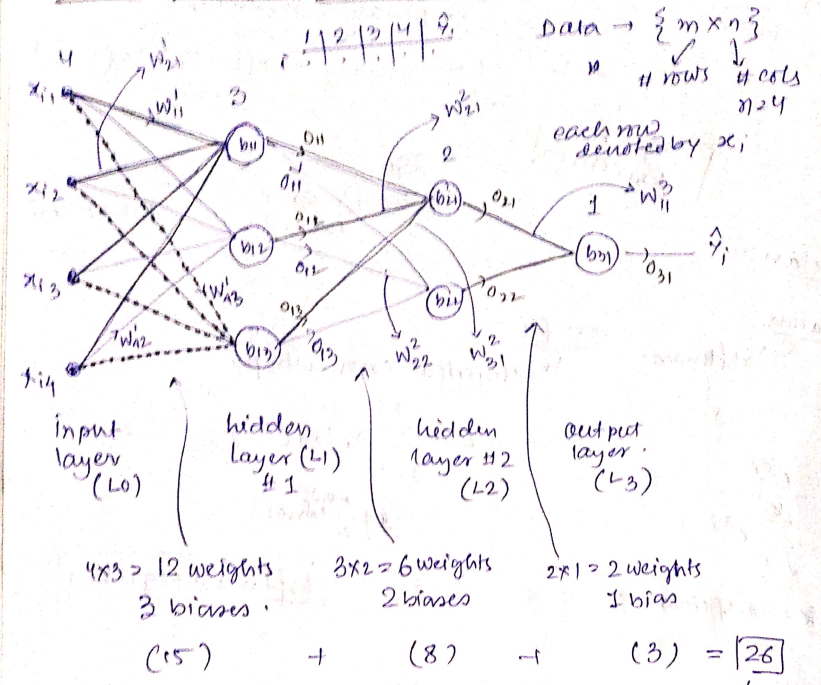
$$f = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$$L = \sum_{j=1}^M Y_j \log(\hat{Y}_j)$$
 (4 classes) $\rightarrow$ softmax Regression
 K = no. of classes

Regression.
 Activation $\rightarrow$ Linear
 loss fn. $\rightarrow$ mse $\rightarrow (Y_i - \hat{Y}_i)^2$
Linear Regression

loss fn	Activation	o/p
Hinge Loss	Sep.	perceptron $\rightarrow$ binary classifier -1, 1
Log loss (binary cross entropy)	Sigmoid	logistic Regression (0-1) $\rightarrow$ binary classifier
Categorical cross entropy	Softmax	softmax regression $\rightarrow$ multi class classification
mse	Linear	linear Regression $\rightarrow$ number

MLP Notation (Multi Layer Perceptron)



in the time of training the backpropagation algorithm try to solve 26 weights & biases.

